

# Dhirendra Pratap Singh

Senior Backend Engineer | Distributed Systems | Java | Kafka | Cloud | AI - Integrated Tooling  
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## RIPPLER

 AI Powered - Code Impact Analysis Platform

## SKILLS

### PRIVATE CLOUD

Self-hosted infrastructure running 32+ services with Advanced Tooling

Docker • Nginx • CI/CD

### LANGUAGES

Over 5000: lines

Java • Python • TypeScript

### FRAMEWORKS

Over 2000: lines

Apache Kafka • ReactJS • Spring Boot

Shedlock • NextJS • Shell Scripting

### TOOLS/PLATFORMS

GCP • OCI • Microsoft Flow

PostgreSQL • MySQL • Git • Jenkins

## CERTIFICATION LINKS

Arcade:Google | HackerRank:Certificates

| freeCodeCamp:300Hrs UI | DSA |

GCP:Core Infra | Leetcode: 1005 |

Oracle:Foundations Associate

## EDUCATION

Magna Cum Laude

CGPA: 8.62 / 10.0

### DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY

BTECH IN ELECTRONICS &

COMMUNICATION

Sep 2020 | Lucknow, India

Babu Banarsi Das National Institute of Technology & Management

### CENTRAL ACADEMY SCHOOLS

Grad. May 2015 | Lucknow, India

## COURSEWORK

### GRADUATE

Cloud Computing | Machine Learning |

Open Source Software Engineering | AI

Integrations | Object-Oriented

Programming

### UNDERGRADUATE

Operating Systems | Artificial Intelligence

+ Practicum | Unix Tools and Scripting |

Information Retrieval

## WORK EXPERIENCE

Total Experience: 5.7 years

### CITI | FULL STACK JAVA DEVELOPER (ASSISTANT MANAGER)

July 2022 – Present | Pune, Maharashtra, India

- Designed and developed **Rippler** , a distributed AI-powered PR impact analysis platform for microservice ecosystems, enabling **real-time LLM-based code change evaluation and automated downstream dependency detection**, reducing manual impact analysis effort and improving production risk visibility across services.
- Contributed to a core-stream within **Citi Commercial Banking (CCB)**, engineering a distributed integration layer that unified CRM, digital banking, authentication, and treasury systems to automate user onboarding, streamline cross-platform data flows, and improve end-to-end client experience.
- Architected high-performance cache layers using **Hazelcast distributed clusters**, reducing critical API latency to <10ms and improving failover resilience under peak loads with main services as fallback.
- Designed a Kafka-based event pipeline to replicate database events across microservices, improving system resilience and enabling reliable cross-service synchronization.
- Implemented **circuit-breaking, retry, and fault-tolerance mechanisms** using **Resilience4j**, preventing cascading failures and improving system stability during production incidents.
- Engineered distributed scheduled job coordination using **ShedLock**, ensuring single-execution guarantees across horizontally scaled services, along with externalized configuration to handle job management from outside the system.
- Architected centralized configuration service enabling **configuration-driven microservice behaviour** across teams, supporting dynamic runtime updates using Spring Cloud @RefreshScope.
- Spearheaded **Trade Commercial E-loans digitization**, driving measurable digital adoption and streamlining customer onboarding workflows.
- Led VM-to-cloud migration efforts with auto-scaling and load-based billing optimizations, reducing infrastructure costs and improving deployment agility.
- Built and launched **UserManagement Pulse**, a full-stack internal product (React, Node, Java microservices), independently owning end-to-end lifecycle from architecture to CI/CD deployment.
- Documented system architecture and operational playbooks using **Docusaurus**, improving debugging efficiency and cross-team self-service enablement.

### CAPGEMINI | SENIOR SOFTWARE ENGINEER

Sep 2020 – Jun 2022 | Pune, Maharashtra, India

- Developed a scalable automated incident processor (800 lines of TACL *Tandem Advanced Command Language* used in DebitCard/Credit Card Txns) for **Discover Financial Services**,  **auto-closing 98% of incidents**, cutting dwell time to net zero. Full-stack contributions in ReactJS, Spring Boot, and PostgreSQL.

### SUVASAA | REMOTE SOFTWARE ENGINEERING INTERN

May 2020 – Aug 2020 | Atlanta, GA, USA

- Full-stack development of an e-commerce platform using ReactJS and Firebase. Integrating online payments via Stripe API and managed segregated customer/admin routes.

## PERSONAL PROJECTS

**PRIVATE CLOUD 32+ SERVICES | FAANG-CRACKER | GITHUB RESUME` PROJECT | CODEFORCES LAMP | AUTO-COMMITTER**